# Python Lab + Assessment

**Duration:** 90 Minutes **Total Marks:** 100

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- **Core Python (Problems 1–10)**

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**Problem 1**

**Question:** Find the second largest element from a list without using built-in sort
**Input data:** A list of integers
**Output data:** Second largest element

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**Problem 2**

**Question:** Count frequency of each character in a string using dictionary
**Input data:** A string
**Output data:** Dictionary with character count

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**Problem 3**

**Question:** Find common elements between two lists without using set
**Input data:** Two lists
**Output data:** List of common elements

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**Problem 4**

**Question:** Check whether a number is Armstrong or not
**Input data:** Integer n
**Output data:** True / False

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**Problem 5**

**Question:** Sort a list of tuples based on marks using lambda
**Input data:** List of tuples [(name, marks)]
**Output data:** Sorted list

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**Problem 6**

**Question:** Find the first non-repeating character in a string
**Input data:** A string
**Output data:** Character

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**Problem 7**

**Question:** Remove duplicates from a list while maintaining order
**Input data:** List of numbers
**Output data:** List without duplicates

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**Problem 8**

**Question:** Swap keys and values in a dictionary
**Input data:** Dictionary
**Output data:** Swapped dictionary

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**Problem 9**

**Question:** Find all pairs in a list whose sum equals a target value
**Input data:** List of integers, target value
**Output data:** List of pairs

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**Problem 10**

**Question:** Flatten a nested list
**Input data:** Nested list
**Output data:** Flattened list

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- **NumPy (Problems 11–18)**

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**Problem 11**

**Question:** Find mean, median, and standard deviation of an array
**Input data:** NumPy array
**Output data:** Mean, median, std deviation

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**Problem 12**

**Question:** Perform element-wise multiplication of two arrays
**Input data:** Two NumPy arrays
**Output data:** Resultant array

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**Problem 13**

**Question:** Transpose a 2D matrix
**Input data:** 2D NumPy array
**Output data:** Transposed matrix

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**Problem 14**

**Question:** Replace all odd numbers in an array with -1
**Input data:** NumPy array
**Output data:** Modified array

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**Problem 15**

**Question:** Extract diagonal elements from a matrix
**Input data:** 2D NumPy array
**Output data:** Diagonal elements

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**Problem 16**

**Question:** Normalize values of an array between 0 and 1
**Input data:** NumPy array
**Output data:** Normalized array

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**Problem 17**

**Question:** Perform matrix multiplication
**Input data:** Two matrices
**Output data:** Result matrix

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**Problem 18**

**Question:** Find indices of elements greater than a given value
**Input data:** NumPy array, value
**Output data:** Indices

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- **Matplotlib (Problems 19–25)**

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**Problem 19**

**Question:** Create a line plot using given x and y values
**Input data:** Two lists (x, y)
**Output data:** Line plot

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**Problem 20**

**Question:** Create a bar chart for product sales
**Input data:** Sales data of 5 products
**Output data:** Bar chart

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**Problem 21**

**Question:** Create a pie chart showing percentage distribution
**Input data:** Category-wise data
**Output data:** Pie chart

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**Problem 22**

**Question:** Plot a histogram for a dataset
**Input data:** Random dataset
**Output data:** Histogram

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**Problem 23**

**Question:** Plot two datasets on the same graph with legend
**Input data:** Two datasets
**Output data:** Combined plot

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**Problem 24**

**Question:** Create a scatter plot and highlight the maximum point
**Input data:** Data points
**Output data:** Scatter plot

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**Problem 25**

**Question:** Customize a plot with title, labels, and grid
**Input data:** Monthly revenue data
**Output data:** Styled plot